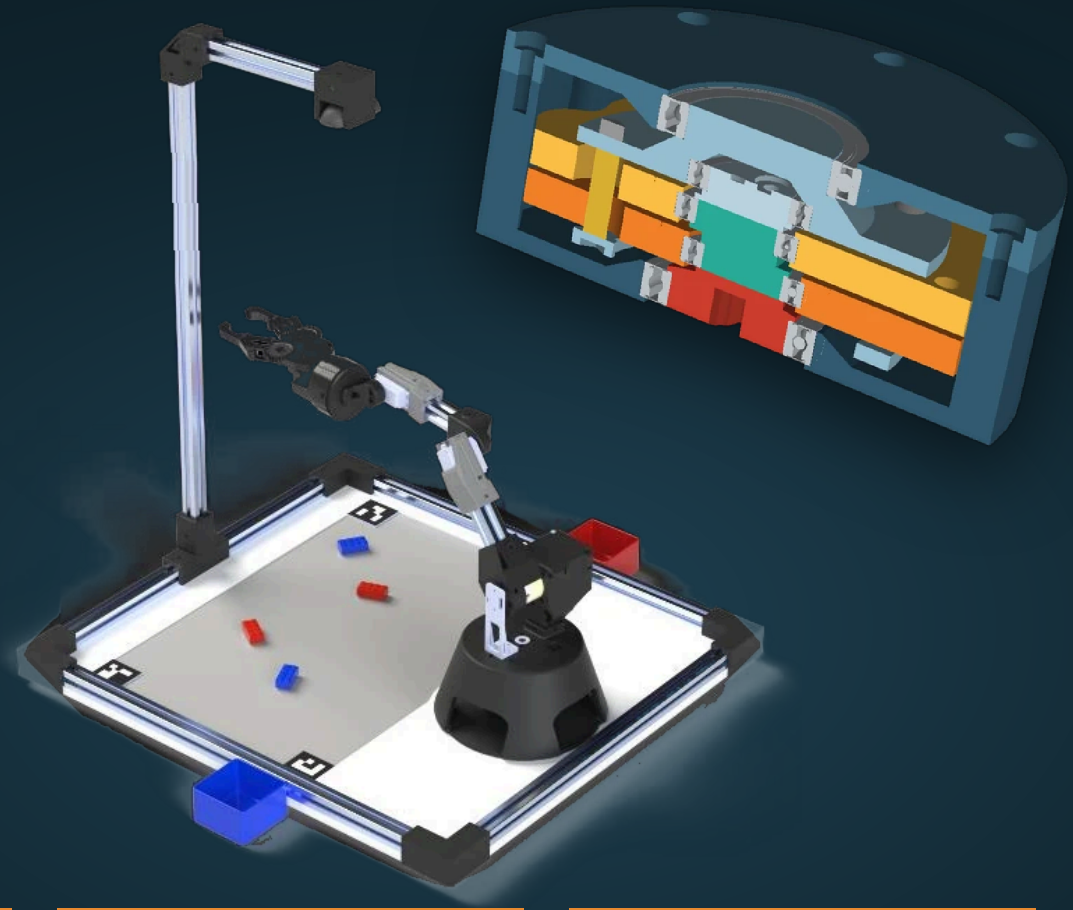
ENGINEERING PORTFOLIO

# Jamal Charkie

Mechanical engineering student at DTU (BEng, 5th semester) and teaching assistant, focused on mechanical design and mechatronics.

I like taking an idea all the way: **calculating it, designing it in SolidWorks and building a prototype** that gets tested and used.



OPEN TO

An engineering internship from February 2027, and a student job alongside my studies.

01 · DTU

**ARES-4 modular robot arm**

**48 / 50**

bricks sorted, fully automatic

02 · INDEPENDENT

**Cycloidal QDD actuator**

**15 : 1**

design ratio · in progress, CAD complete

03 · ASSOCIATION

**Wall-mounted QR scanner**

**V2**

in use every week

04 · DTU

**Motor-driven syringe pump**

**99–101 %**

of the set flow

**Also built:** a vision-guided nerf turret (YOLOv8n, steppers) · a bike-powered phone charger · a table football goalkeeper bot.

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3D models and video: **jamalch15.github.io/portfolio**

01

COMPLETED DTU · team of 3

# ARES-4: a modular robot arm, built for many tasks

Its tools change by hand with a twist, and a browser dashboard sets up and runs new jobs. Tested on **fully automatic LEGO sorting**: find a brick, pick it, drop it in its colour's bin, repeat until the table is clear.

| DESIGN | ELECTRONICS | SOFTWARE |
|---|---|---|
| SolidWorks · Maple | KiCad · ESP32-S3 | Python · FastAPI · vision |

**48 / 50**
bricks sorted with no one touching it

**≈500 g**
payload carried

**≈7 Nm**
shoulder peak, for ≈4 Nm needed

## WHAT I DID

In a team of three (DTU, 2026), I drove the **mechanical design, the electronics and the software**: motor sizing and CAD, KiCad wiring, ESP32-S3 firmware, kinematics and the control dashboard.

## DESIGN DECISIONS

**Strong enough at full reach.** About 4 Nm needed at the shoulder; a 27:1 geared stepper gives about 7 Nm peak, confirmed by our torque test.

**Light up top.** Servos at the elbow and wrist keep the mass near the base.

**Quick-swap tools.** Gripper or electromagnet, changed by hand without tools; a sideways pogo connector powers the tool when it locks.

**Encoder-corrected shoulder.** An AS5048A measures the real angle after each move.

**Known limits.** Backlash limits precision; next step: closed-loop control on every joint.

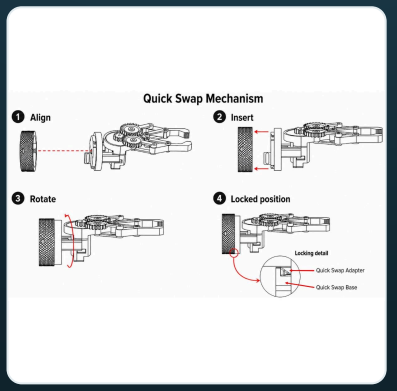
Jamal Charkie · 01 ARES-4 · code, KiCad and calculations: github.com/Jamalch15/ARES-4



THE ARM IN 3D, FROM THE ORIGINAL SOLIDWORKS ASSEMBLY



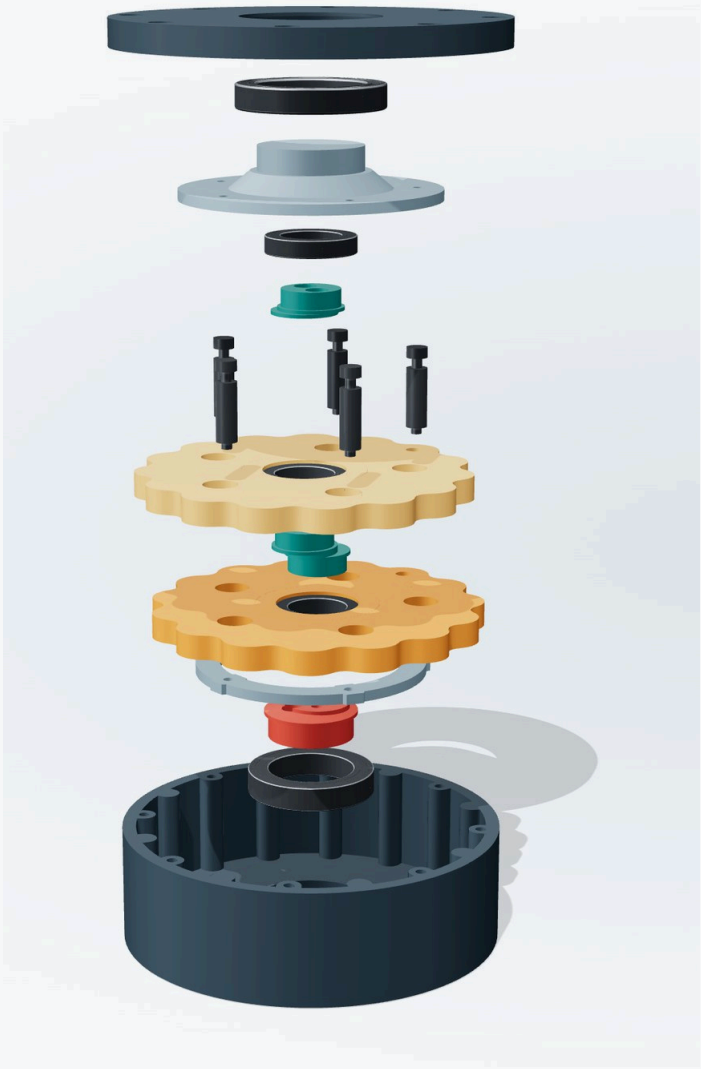
THE REAL CELL




QUICK-SWAP: ALIGN, INSERT, ROTATE, LOCK



DASHBOARD: GHOST ARM AND PLANNED PATH



02

IN PROGRESS · CAD COMPLETE Independent project

# A printable robot joint, designed to be backdrivable

A low-ratio cycloidal drive on a drone motor: the quasi-direct-drive kind used in legged robots, **designed to be printed at home** around a few bought parts.

| DESIGN | DRIVE | BUILD |
|---|---|---|
| SolidWorks · cycloids | BLDC · FOC (planned) | FDM printing |

**15 : 1**
reduction, in a 12–16:1 target

**≈16 Nm**
peak torque, design target

**≈470 rpm**
top speed, design target

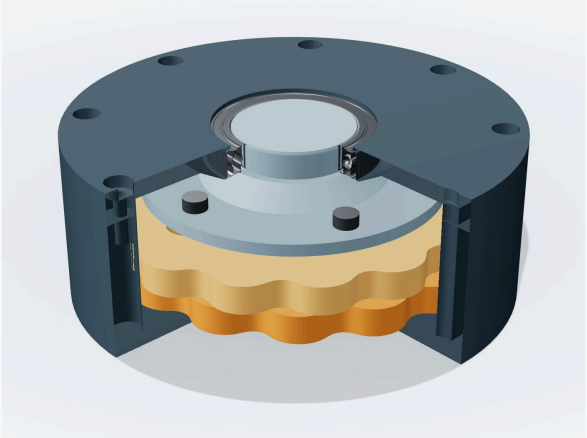
## DESIGN DECISIONS

**Low ratio on purpose.** The aim: a joint that stays backdrivable and quick to respond.

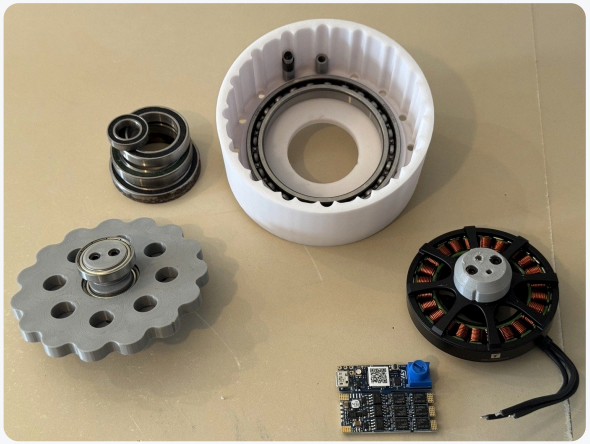
**Drone motor, not a stepper.** Sized for a T-Motor MN6007 II outrunner, with field-oriented control planned.

**No ring pins.** The discs roll on a profile in the printed housing; printed sleeves over screws had caused seam friction.

**Serviceable shaft.** A three-piece eccentric shaft clamped with M3 screws, so the bearings fit inside the discs.



CUTAWAY: TWO DISCS INSIDE THE LOBED HOUSING



FIRST PROTOTYPE: PRINTED AND FITTED WITH REAL BEARINGS

**Next:** fit samples, assembly and hand-turn tests of this design.

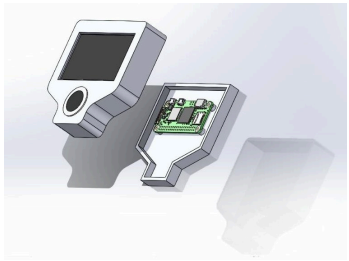
03

IN USE EVERY WEEK · Independent, for an association

# QR check-in terminal: prototype, field evaluation, redesign

I designed and built the hardware: the enclosure and wall mount around the display, GM60 scanner and Raspberry Pi. The app software was handled separately.

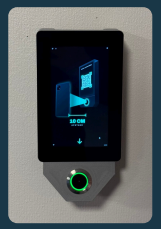
V1 · PROTOTYPE



Field evaluation: **issues and real-world use logged** on the wall.

→

V2 · REDESIGN, IN SERVICE



- Larger official 5-inch Raspberry Pi display
- A clearly separate scan opening
- An easier-to-read front
- Simpler wall installation, sturdier mounting
- Quick to assemble and service

**In use at the association every week.**

## DESIGN DECISIONS

**Designed around the real parts.** The display, scanner and Raspberry Pi set the packaging; the CAD covers fasteners, clearances, cable routing and assembly access.

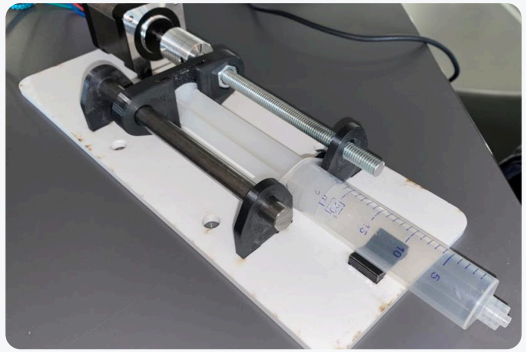
**Print-ready.** Every housing part is designed for FDM, then printed, assembled and tested on the real hardware.

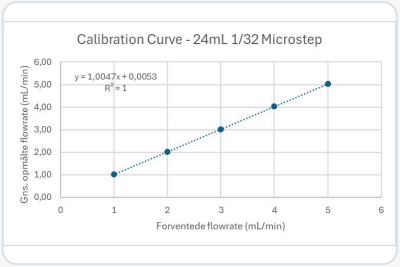
04

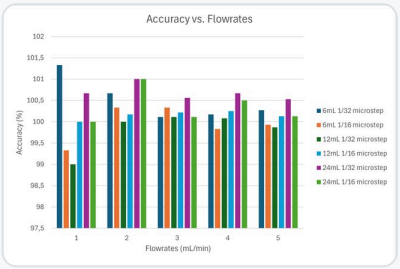
COMPLETED · DTU course project · team of 3

# A printed syringe pump within 99–101 % of set flow

A modular pump that pushes and pulls, takes 6–24 mL syringes and turns stepper rotation into precise plunger travel. **I modelled it, built it and measured it.**






**99–101 %**
of set flow, 6, 12 and 24 mL syringes

**1–5**
mL/min tested, 3 repeats per point

**0.01 g**
scale for every measurement

## WHAT I DID

**CAD modelling** (with feedback from the group), **building the prototype**, and the **data collection and processing**.